

C STATISTICAL DETAILS

Recall and FAR bounds are one-sided 95% Clopper-Pearson bounds. The two paired 95% confidence intervals in Table 1 use the bootstrap distribution known in the E1 setting. The sign-dig permutation results were computed with 20,000 one-sided draws and correct dimensionality, i.e., the correct empirical values for α , which we report as 2×10^{-4} rather than as mathematically prohibitive zero. Exact equality statements for gate, best-val, and early-stop come from recordwise equality of both stored order loss and stored error accuracy, not a null test.